

Chapter 10 – Task Scheduler, Temp Files, Event Viewer & Control Panel

1. Task Scheduler (Automation Tool)

◆ What is Task Scheduler?

Task Scheduler is a built-in Windows tool that allows you to **automatically run programs, scripts, or tasks** at a specific time or event.

👉 Simple Meaning:

It works like an **alarm system for tasks** 

💡 Example:

- Run antivirus scan every day at 10 PM 
- Open Chrome automatically when system starts 

Where is it used in IT?

- System maintenance (cleanup, updates)
- Backup automation
- Security monitoring
- Malware persistence ( attackers misuse this)

How to Open Task Scheduler

Windows + R → taskschd.msc

Step-by-Step: Create Task

1 Open Task Scheduler

2 Click **Create Task**

3 Enter:

- Name → Chrome Browser
- Description → Open Chrome browser

4 Choose:

- ✓ Run only when user is logged on



Trigger (When to run?)

- One time
- Daily
- At startup
- On login



Action (What to run?)

- Click **New**
- Select → Start a Program
- Add path:

C:\Program Files\Google\Chrome\Application\chrome.exe



Click OK → Task Created 



Security Insight (VERY IMPORTANT)



If any task has path like:

C:\Users\...\AppData\...



It can be MALICIOUS (virus)

Malware Behavior (Real IT Scenario)

How Ransomware Works:

1. User downloads file (Photoshop etc.)
2. Installer extracts files in **Temp / AppData**
3. Malicious script runs silently
4. Files get encrypted 
5. Attacker demands money 

Where malware hides:

- AppData folder 
- Run / RunOnce registry 
- Task Scheduler 
- Temp folders 

Key Terms:

- **Dropper** → Malware that installs other malware
- **PUP** → Potentially Unwanted Program

2. Temp & %TEMP% Folder (VERY IMPORTANT)

What is Temp Folder?

Temporary files storage used by:

- Windows
- Applications
- Installers

 Locations:

 System Temp:

C:\Windows\Temp

 User Temp (%TEMP%):

C:\Users\<Username>\AppData\Local\Temp

 Open quickly:

Windows + R → %TEMP%

 What gets stored?

Type	Example
Cache	Chrome, Office
Installer files	Setup files
Logs	Crash reports
Print data	Print spool files

 Daily Life Example:

Like **kitchen waste bin** 

→ Temporary use → then throw away

 IT Support Usage:

- Check virus entry point
 - Recover deleted installation traces
 - Analyze user activity
-

💡 **Important Difference:**

- **Temp (Windows)** → System files
 - **%TEMP% (User)** → App/3rd party files
-

🧪 **3. Virus Investigation (Real Scenario)**

🎯 **Situation:**

User says: "Virus came automatically"

💻 **IT Answer:**

- 👉 Virus NEVER comes automatically
 - 👉 It comes during:
 - Software installation
 - File download
-

🔍 **How to prove?**

Check:

1. Temp Folder

- Installation logs present

2. AppData Folder

- Application traces

3. Downloads Folder

- Original file
-

🧠 **Example:**

User deletes:

- Setup file ❌
- Installed program ❌

👉 BUT still found in:

- %TEMP% ✓
- AppData ✓

4. Event Viewer (System Logs)

◆ What is Event Viewer?

A tool that shows:

- System logs
- Errors
- Warnings
- Security logs

Open:

Windows + R → eventvwr.msc

Used For:

- Troubleshooting crashes
- Finding malware activity
- Tracking login attempts

◆ What it does

- Records **everything happening in Windows**
- Errors, warnings, logins, crashes

◆ Log Types

- **Critical** ✖ – System crash, sudden shutdown
 - **Error** ⚠ – Something failed
 - **Warning** ⚠ – Potential problem
 - **Information** ⓘ – Normal activity
 - **Verbose** 🔍 – Detailed debug info
-

Event Viewer Log File Location

◆ Path

C:\Windows\System32\winevt\Logs

◆ File type

.evtx

Used in:

- Troubleshooting
 - Forensics
 - Security investigations
-

Blue Screen (BSOD) Logs

◆ MiniDump Path

C:\Windows\Minidump

◆ Full Memory Dump

C:\Windows\memory.dmp

◆ Difference

- **MiniDump** → OS error details
- **Memory Dump** → RAM crash, memory overflow

⚙️ **Enable BSOD Logging (If Logs Not Generating)**

1. Windows + R → sysdm.cpl
2. Advanced → Startup & Recovery → Settings
3. Tick ☒ **Write an event to the system log**

Application Logs

◆ **Location**

Event Viewer → Windows Logs → Application

◆ **What logs here**

- App install errors
- Game crashes
- Software faults

Security Logs

◆ **Location**

Event Viewer → Windows Logs → Security

◆ **What it shows**

- Login time
- Logout time
- Failed login attempts
- Account changes

Used by:

- IT admins
 - Security teams
 - Forensics
-

Critical Error Example

◆ Scenario

Laptop shuts down suddenly

◆ Log Generated

Critical Error in Event Viewer → System

Indicates:

- Power failure
- Overheating
- Hardware fault

5. Control Panel (Complete Overview)

◆ What is Control Panel?

Central place to manage:

- System settings
- Devices
- Security
- User accounts

Important Options:

Credential Manager

Stores:

- App login credentials
 - Windows credentials
-

Concept:

- Password → Stored normally
 - PIN → Certificate-based 
-

Problem Scenario:

 "Credential expired" error

Solution:

1. Open Credential Manager
 2. Delete old credentials
 3. Restart system
 4. Setup PIN again
-

Date & Time

- Add multiple clocks 
 - Internet time sync (used in army, airports)
-

Device Encryption (BitLocker)

What is BitLocker?

Microsoft tool to encrypt disk data 

Features:

- Data becomes unreadable without key
 - Protects against disk theft
-

Recovery Key Options:

- Save to Microsoft account
 - Save to file
 - Print
-

Access Recovery Key:

<https://aka.ms/recoverykey>

Important:

- BIOS update → may ask recovery key
 - Always backup key
-

Compliance Rules (Corporate IT)

- ✓ BIOS updated
- ✓ Secure Boot ON
- ✓ Disk encrypted

Otherwise:

 Device becomes **non-compliant**

Devices and Printers

- Manage:
 - Keyboard
 - Mouse
 - Printer
 - Bluetooth devices
-

Ease of Access Center

For visually impaired users 

Tools:

- Narrator 
 - Magnifier 
-

Final IT Support Troubleshooting Tips

Quick Problem Solving:

System Slow:

- Check Task Manager
 - Disable startup apps
 - Update drivers
-

WiFi Slow:

- Check if issue is:
 - Single user → Driver issue
 - Multiple users → Network issue
-

Virus Detection:

Check:

- Task Scheduler
 - AppData
 - Run registry
 - Temp folder
-

Login / Credential Issue:

- Reset credentials
 - Recreate PIN
-

Real-Life Summary

Concept	Real-Life Example
Task Scheduler	Alarm clock 
Temp Folder	Dustbin 
BitLocker	Locker with key 
Event Viewer	CCTV recording 
AppData malware	Hidden thief 

Q & A

1. TASK SCHEDULER

 **Q1: What is Task Scheduler?**

 **Answer:**

Task Scheduler is a Windows tool that allows us to **automatically run programs, scripts, or commands at a specific time or event** 

 It works like an **alarm system for tasks**

 Example:

- Run antivirus daily at 10 PM 
- Open browser at startup 

 In IT support:

- Used for automation
- Used for maintenance
- Also used by malware (important point )

 **Q2: Where is Task Scheduler used in real IT environments?**

 **Answer:**

Task Scheduler is used in companies for:

-  Automatic backups
-  Disk cleanup tasks
-  Software updates
-  Security scans
-  Malware persistence (attackers misuse it)

? Q3: How do you create a scheduled task?

✓ Answer (Step-by-Step):

1. Press **Windows + R** → **taskschd.msc**
2. Click **Create Task**
3. Enter:
 - Name
 - Description
4. Go to **Triggers tab**
 - Select: Daily / Startup / Login
5. Go to **Actions tab**
 - Select: Start a Program
 - Add program path
6. Click OK **✓**

👉 Task is created successfully

? Q4: How can Task Scheduler be used by malware?

✓ Answer:

Attackers use Task Scheduler to:

- Run malware automatically at startup
- Execute hidden scripts silently
- Maintain persistence (stay active after restart)

⚠ Suspicious sign:

👉 Task path contains:

C:\Users\...\AppData\

👉 Because:

- AppData is hidden
- No admin permission required

💡 This is a **VERY IMPORTANT** interview point

? **Q5: How do you detect malicious scheduled tasks?**

✅ **Answer:**

Steps:

1. Open Task Scheduler
2. Check all tasks
3. Look for:
 - Unknown names ❌
 - Suspicious paths ❌
4. Verify file location
5. Scan file using antivirus or VirusTotal 🧪

👉 If confirmed malicious → disable/delete task

📁 2. TEMP & %TEMP% FOLDER

? **Q6: What is a Temp folder?**

✅ **Answer:**

Temp folder stores **temporary files created by Windows and applications** 🗑️

👉 These files are used for:

- Installation
- Caching

- Logs
- Temporary processing

💡 Example:

Like a **dustbin** — used temporarily then deleted

❓ **Q7: Difference between Temp and %TEMP%?**

✅ **Answer:**

Folder	Meaning
C:\Windows\Temp	System temp files
%TEMP%	User/application temp files

👉 %TEMP% is more important for **malware investigation** 🔥

❓ **Q8: Why is Temp folder important in troubleshooting?**

✅ **Answer:**

Because it helps in:

- Finding installation traces
- Detecting malware entry
- Checking logs
- Recovering temporary data

👉 Even if user deletes software:

✓ traces still exist in Temp folder

❓ **Q9: Can viruses come automatically?**

❌ **Wrong assumption**

✅ **Correct Answer:**

No, viruses NEVER come automatically

👉 They come from:

- Downloaded files
- Cracked software
- Email attachments

🔍 Proof:

Check:

- Temp folder
- AppData
- Downloads

? **Q10: How do you investigate virus entry?**

✓ **Answer (Real IT Approach):**

1. Check Downloads folder
2. Check %TEMP%
3. Check AppData
4. Check installed programs
5. Identify suspicious file
6. Scan using VirusTotal

👉 This shows professional troubleshooting mindset **100**

3. EVENT VIEWER

? **Q11: What is Event Viewer?**

✓ **Answer:**

Event Viewer is a tool that shows **system logs and events** 

👉 It records:

- Errors
- Warnings
- System activity
- Login attempts

💡 Example:

Like a **CCTV recording system** 📺

? **Q12: What are different log types?**

✅ **Answer:**

Type		Meaning
Critical	❌	System crash
Error	⚠️	Failure
Warning	⚠️	Possible issue
Information	ℹ️	Normal
Verbose	🔍	Detailed logs

? **Q13: Where are Event Viewer logs stored?**

✅ **Answer:**

📍 Location:

C:\Windows\System32\winevt\Logs

📄 File type:

.evtx

👉 Used in:

- Troubleshooting

- Forensics
 - Security analysis
-

? Q14: How do you check system crash (BSOD)?

✓ Answer:

Check:

• MiniDump:

C:\Windows\Minidump

• Full dump:

C:\Windows\memory.dmp

👉 MiniDump = OS error

👉 Memory dump = RAM crash

? Q15: How to troubleshoot sudden shutdown?

✓ Answer:

1. Open Event Viewer
 2. Go to:
→ Windows Logs → System
 3. Look for:
 - Critical errors
 4. Identify:
 - Power issue
 - Overheating
 - Hardware fault
-

 **Q16: What are Security Logs used for?**

 **Answer:**

Security logs track:

- Login attempts
- Failed logins
- User activity
- Account changes

 Used by:

- IT Admins
- Security teams
- Cyber forensics

4. CONTROL PANEL

 **Q17: What is Control Panel?**

 **Answer:**

Control Panel is a centralized tool to manage:

- System settings 
- Devices
- Users
- Security

 It is used daily in IT support

? Q18: What is Credential Manager?

✓ Answer:

Credential Manager stores:

- Saved passwords
- Windows credentials
- App login data

🔑 Example:

Like saved login in browser

? Q19: What is “Credential Expired” issue?

✓ Answer:

Cause:

- Old or corrupted credentials

Solution:

1. Open Credential Manager
 2. Delete old credentials
 3. Restart system
 4. Re-login
-

? Q20: What is BitLocker?

✓ Answer:

BitLocker is a **disk encryption feature** 🔑

👉 It protects data from unauthorized access

💡 Example:

Like a **locker with key**

? Q21: What happens if BitLocker disk is moved to another PC?

✓ Answer:

System will ask for:

👉 25-digit recovery key

Reason:

- Hardware ID mismatch
-

? Q22: Why is BitLocker important in companies?

✓ Answer:

- Protects data theft
 - Required for compliance
 - Prevents unauthorized access
-

? Q23: What is Device Encryption?

✓ Answer:

It encrypts disk data so:

- Data becomes unreadable without key
 - Protects laptop theft
-

🧠 TROUBLESHOOTING QUESTIONS

? Q24: System is slow. What will you check?

✓ Answer:

1. Task Manager (CPU/RAM)
2. Startup apps
3. Background processes

4. Temp files

5. Malware

? Q25: How do you detect malware in system?

✓ Answer:

Check:

- Task Scheduler ⚠
- AppData ⚠
- Run registry ⚠
- Temp folder ⚠

👉 These are **common hiding locations**

? Q26: WiFi is slow. What will you do?

✓ Answer:

Ask:

👉 Single user or multiple users?

If single:

- Update driver
- Check adapter

If multiple:

- Network issue
-

? Q27: Printer not working?

✓ Answer:

1. Check Print Spooler service
 2. Restart service
 3. Check driver
 4. Check connection
-

? Q28: User says “virus came automatically”. What will you say?

✓ Answer:

Explain:

👉 Virus never comes automatically

Then investigate:

- Downloads
 - Temp
 - AppData
-

🔍 FINAL SUMMARY

👉 Task Scheduler = Automation tool ⌚

👉 Temp Folder = Temporary storage 🗑️

👉 Event Viewer = System logs 📄

👉 Control Panel = Settings hub ⚙️

👉 BitLocker = Security 🔒
